

Behavior Tree (BT) Design Guidelines

These guidelines specify design requirements for behavior-tree entry conditions, node responsibilities, and command authority.

01

Parent and child conditions

Parent entry conditions shall cover all situations requiring child maneuvers.

1-1 The parent condition shall be satisfied whenever a child maneuver is required.

Example scenario An enemy firing threat is detected before any hit is received, requiring preemptive evasion.

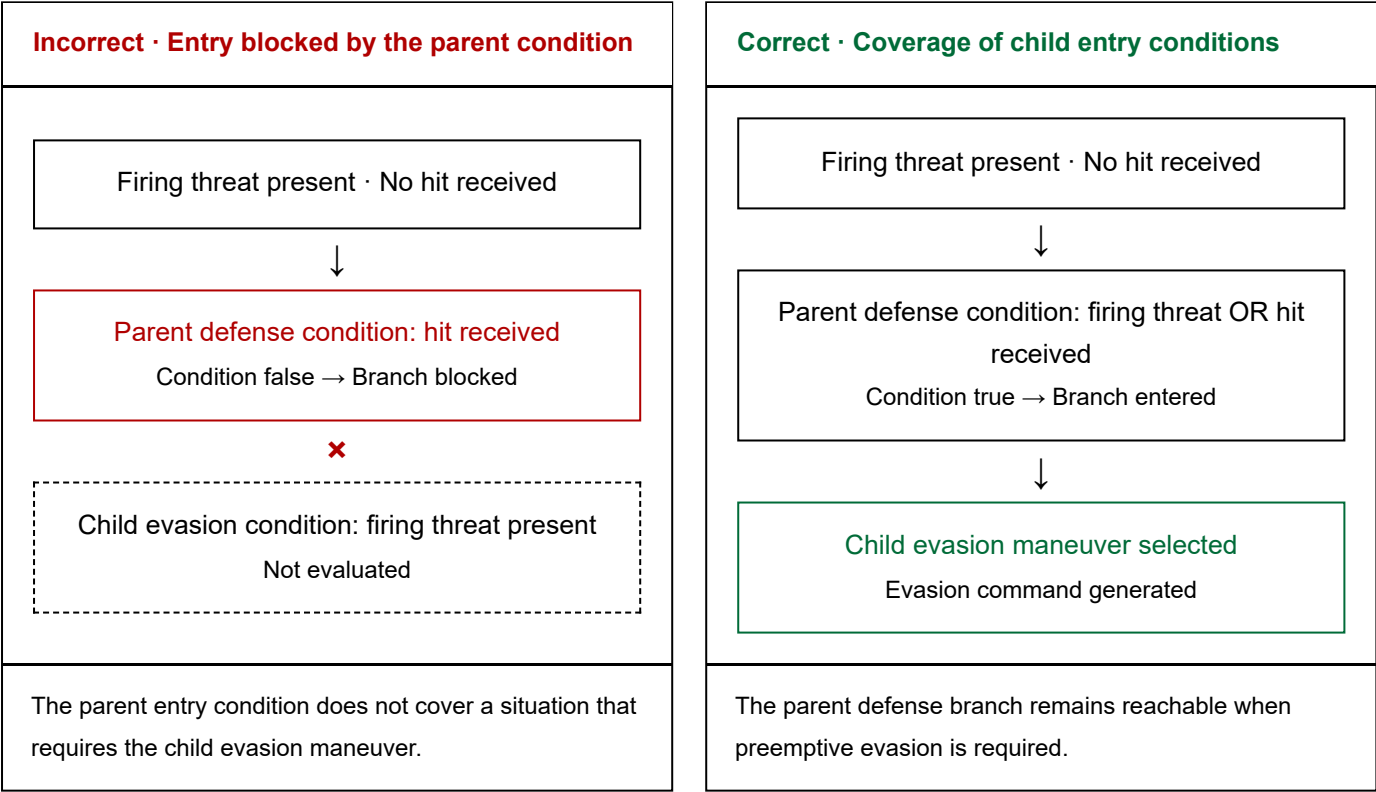


Figure 1-1. Coverage of child entry conditions

1-2 Parent and child nodes shall use a common assessment result to avoid applying different criteria to the same situation.

Example scenario Parent and child nodes assess the same enemy firing threat.

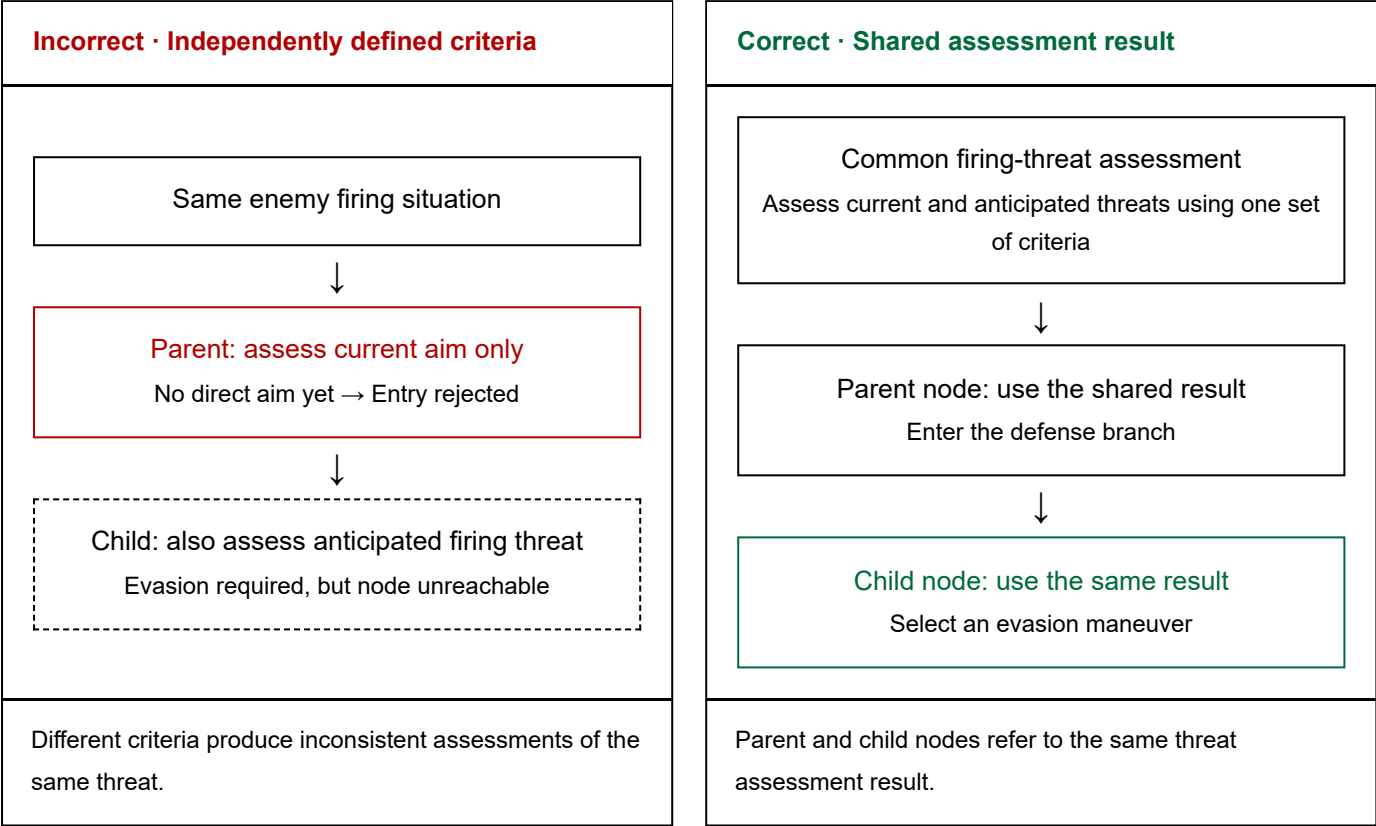


Figure 1-2. Shared assessment result

Gaps in decision-making and control commands shall be prevented.

2-1 For each situation, at least one decision module and one maneuver module shall be selectable.

Example scenario Neither the ownship nor the opponent has a confirmed unilateral advantage.

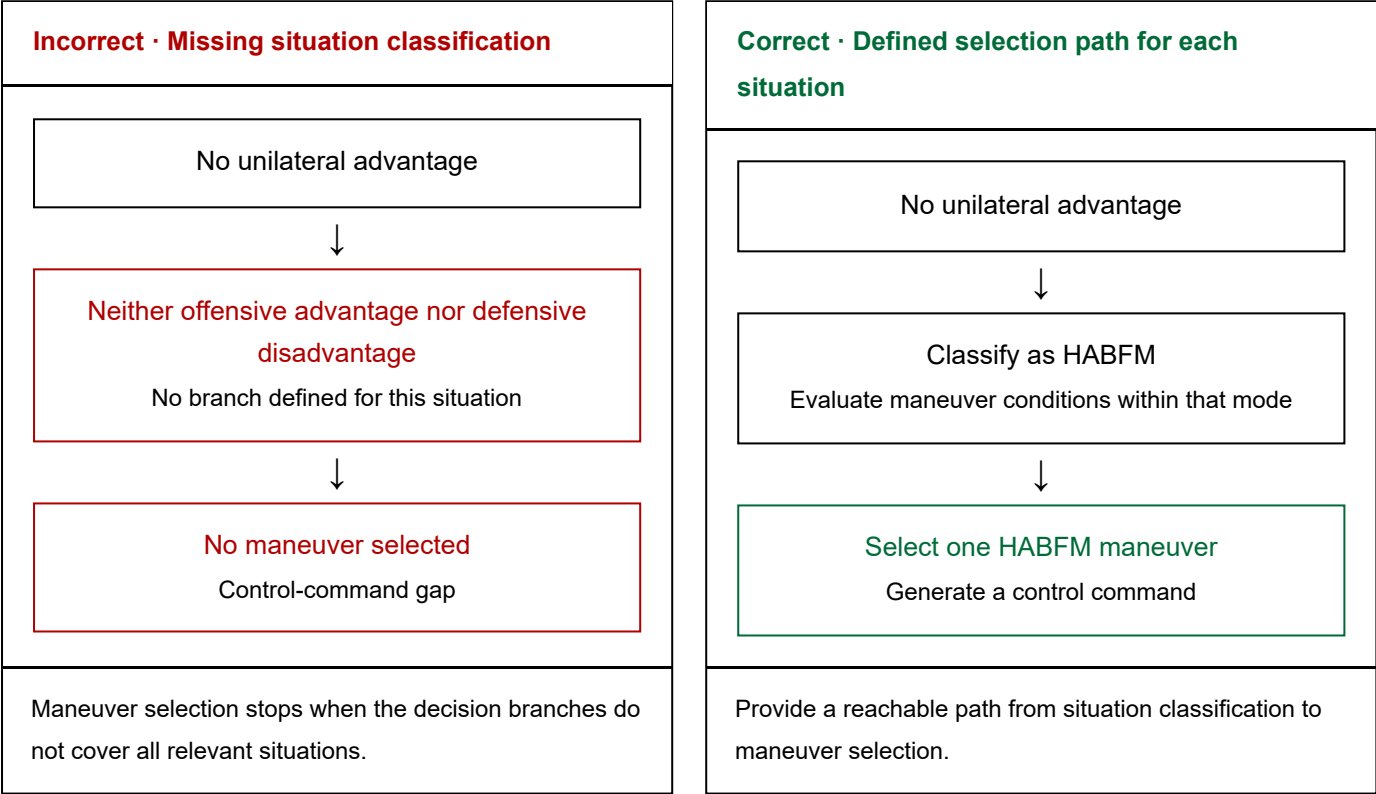


Figure 2-1. Defined selection path for each situation

2-2 **Exception handling shall not be used to bypass defects during development. The causes of exceptions shall be identified and removed through debugging before operational use.**

Example scenario A zero-magnitude error vector is supplied to a direction-error normalization calculation.

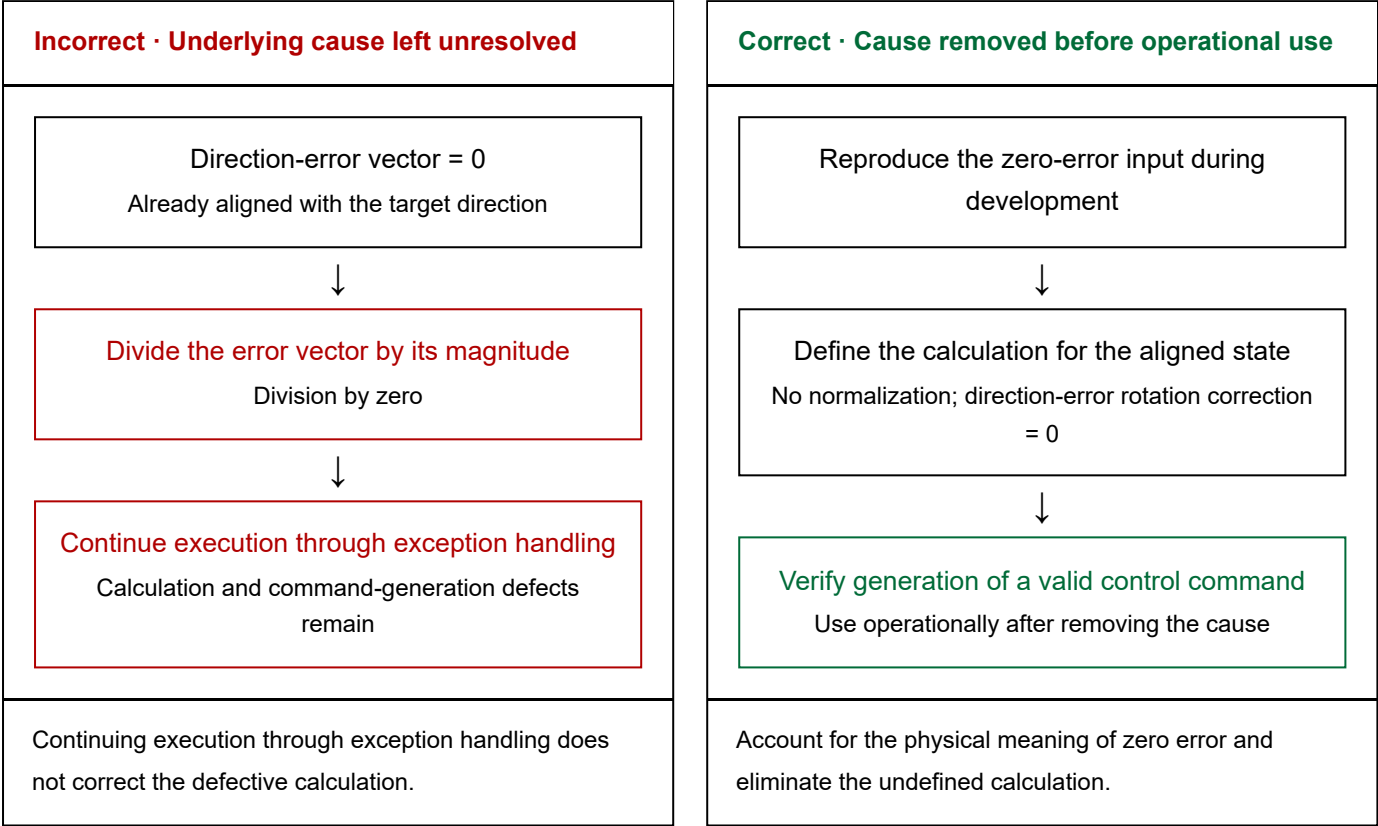


Figure 2-2. Cause removed before operational use

03

Separation of responsibilities

Service, Condition, and Task responsibilities shall be separated and observed.

- 3-1 Services shall perform observation, Conditions shall perform assessment, and Tasks shall generate commands.

Example scenario Observation, assessment, and command-generation functions for a climb maneuver are assigned to BT nodes.

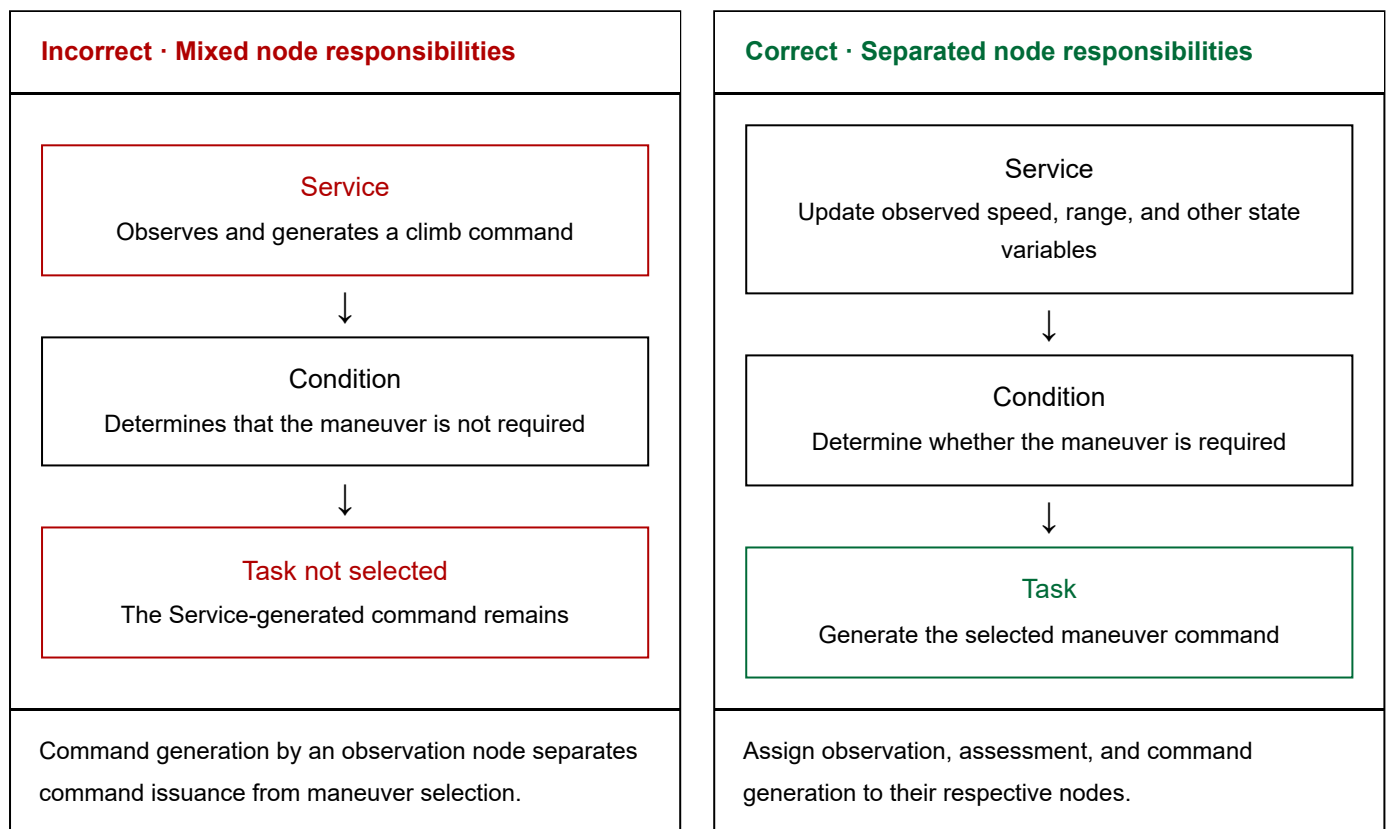


Figure 3-1. Separated node responsibilities

3-2 Observed quantities shall be calculated once and shared.

Example scenario Multiple nodes use the range to the opponent within one control cycle.

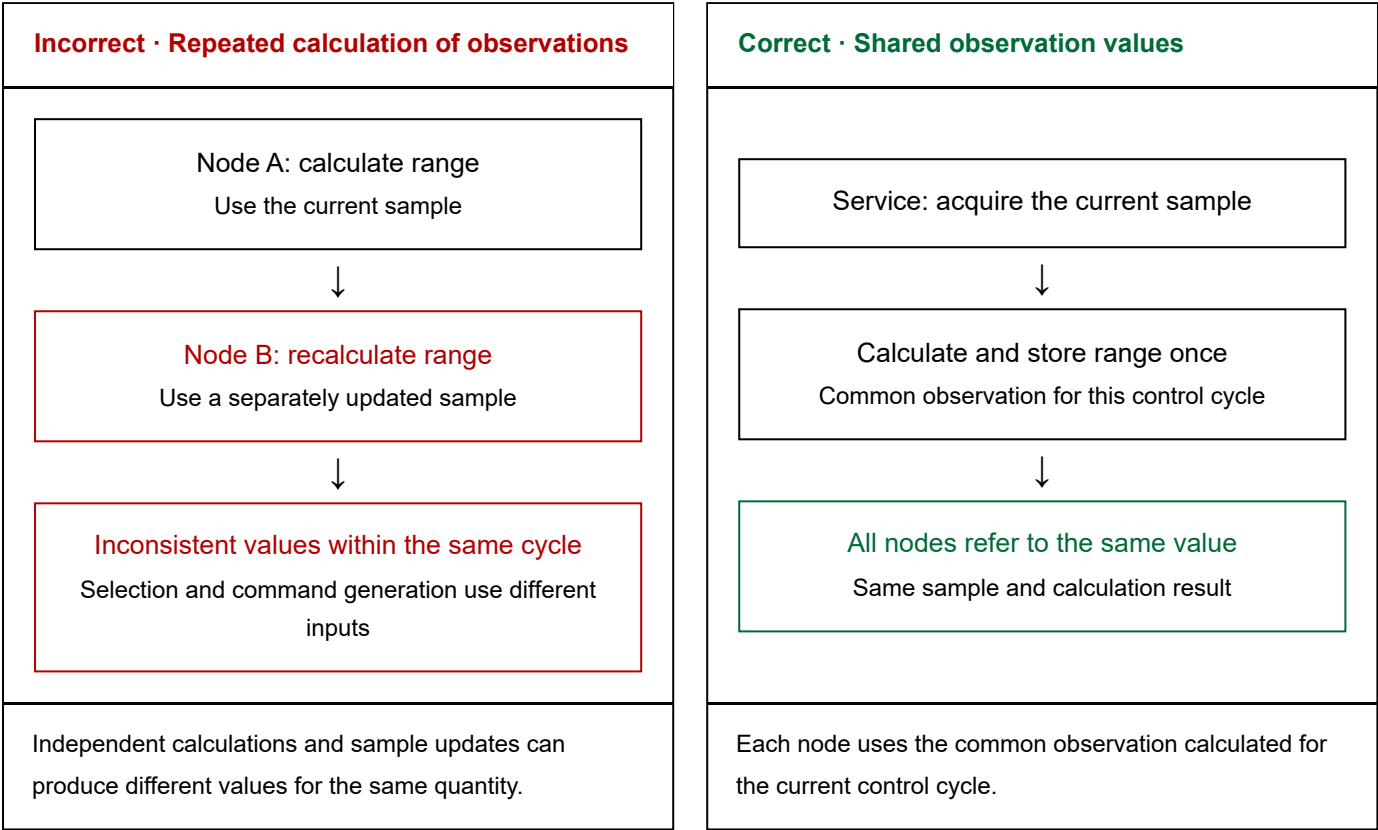


Figure 3-2. Shared observation values

3-3 Conditions and Tasks shall use the same entry assessment result.

Example scenario A Condition approves High-Yo-Yo entry and passes the result to the Task.

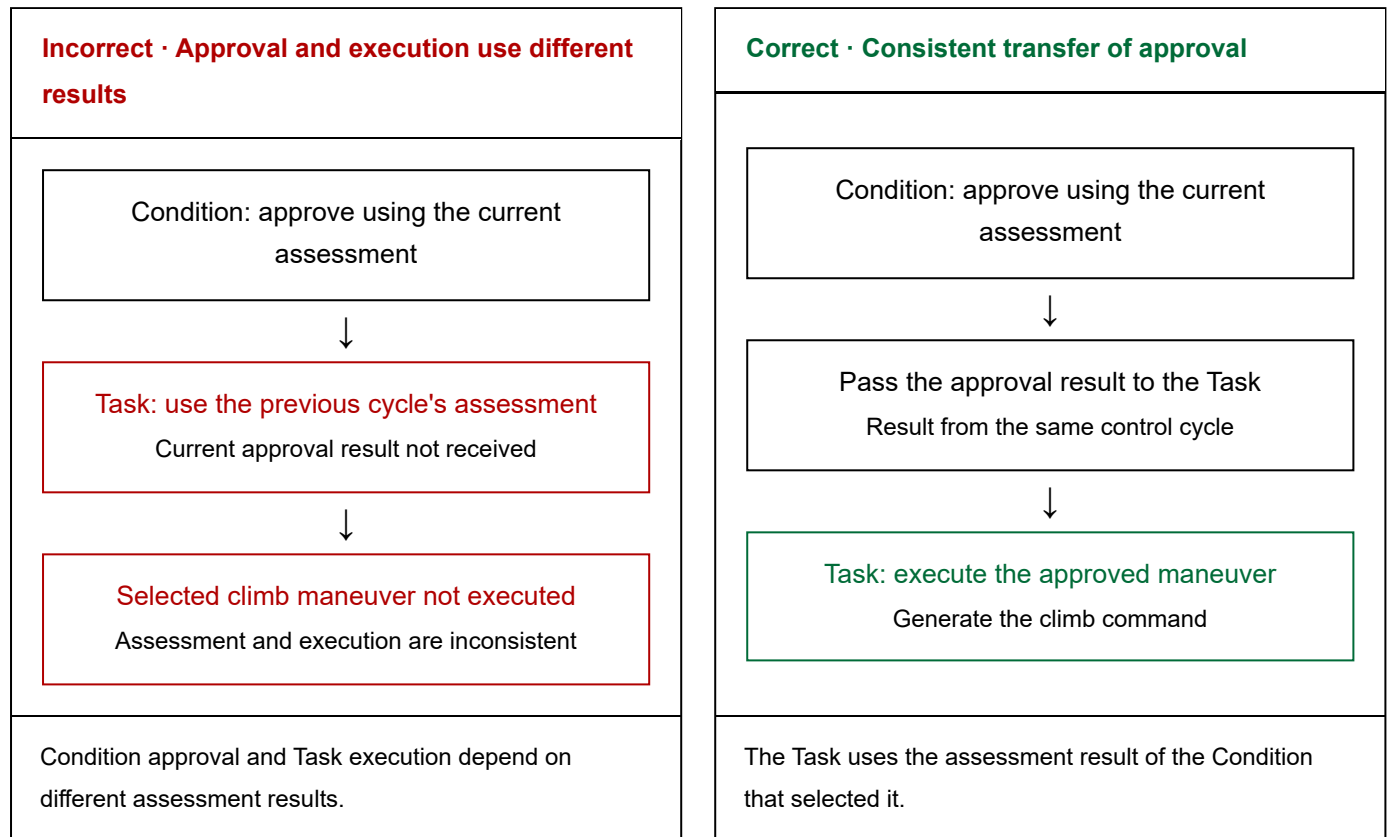
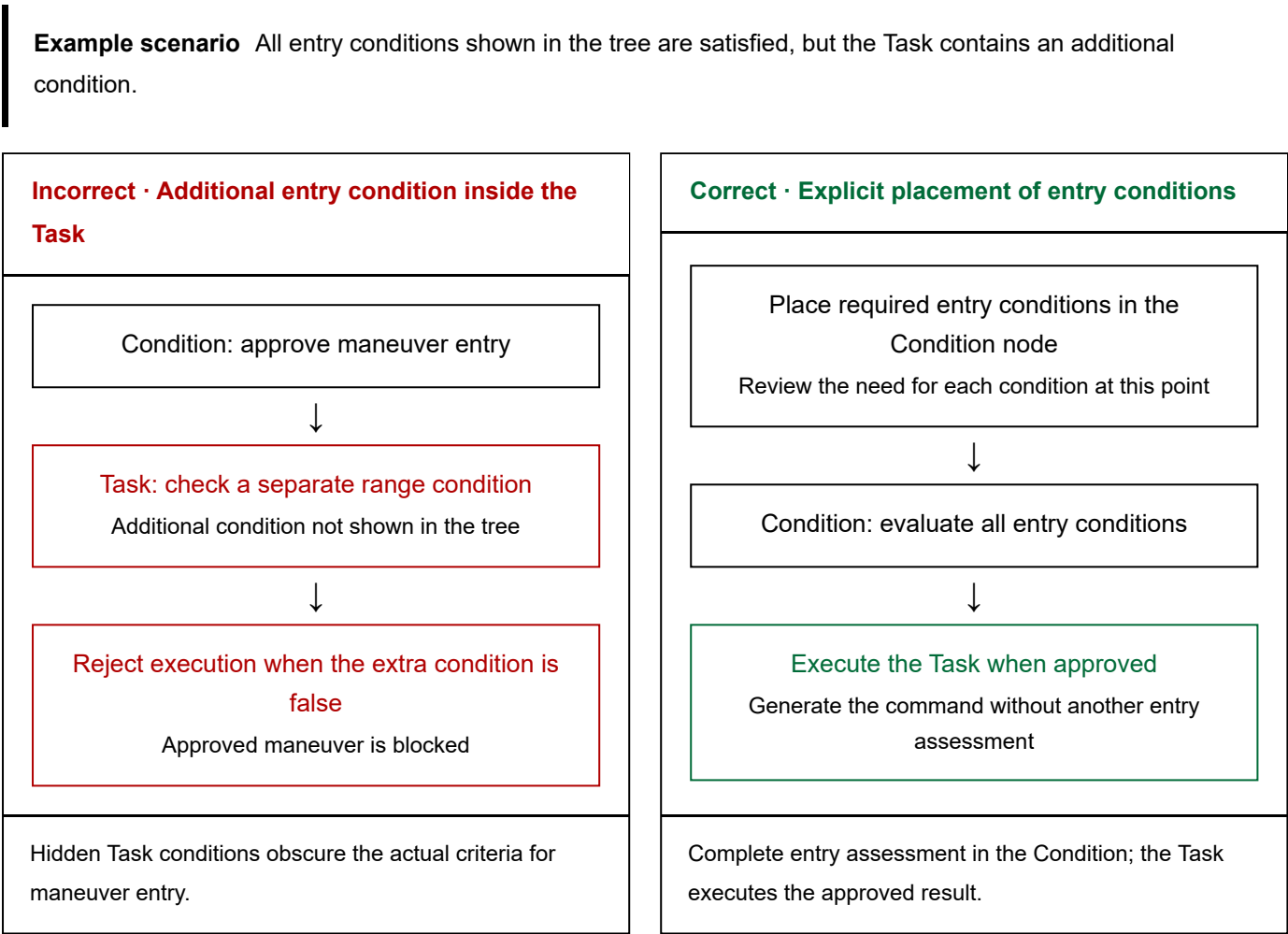
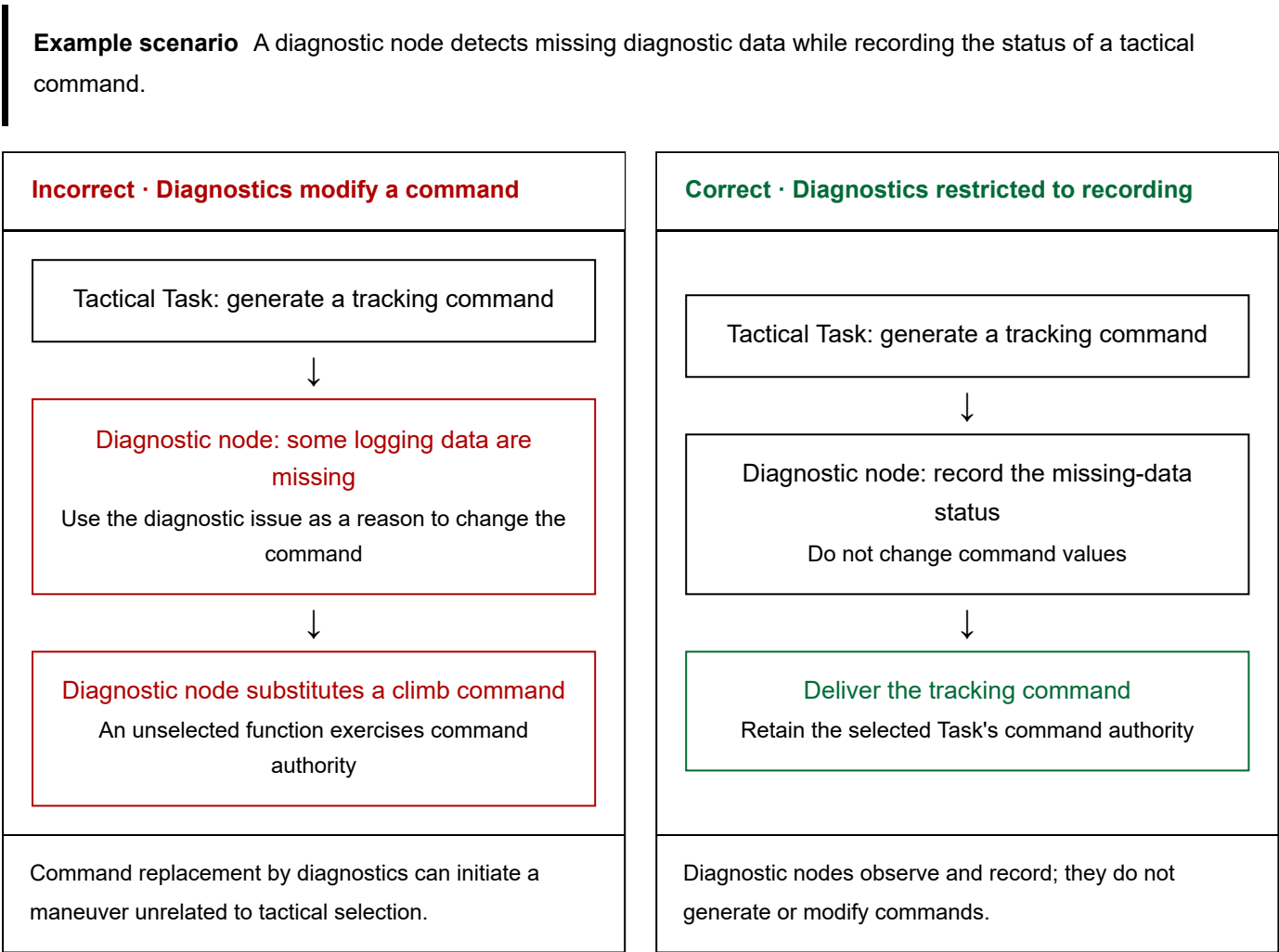


Figure 3-3. Consistent transfer of approval

3-4 Additional entry conditions shall not be introduced inside a Task.



3-5 Diagnostic nodes shall not be assigned command authority.



04

Single command authority

Exactly one module shall hold final command authority.

4-1 Modules with similar entry conditions shall be reviewed for duplication.

Example scenario Two registered evasion modules respond to the same firing threat.

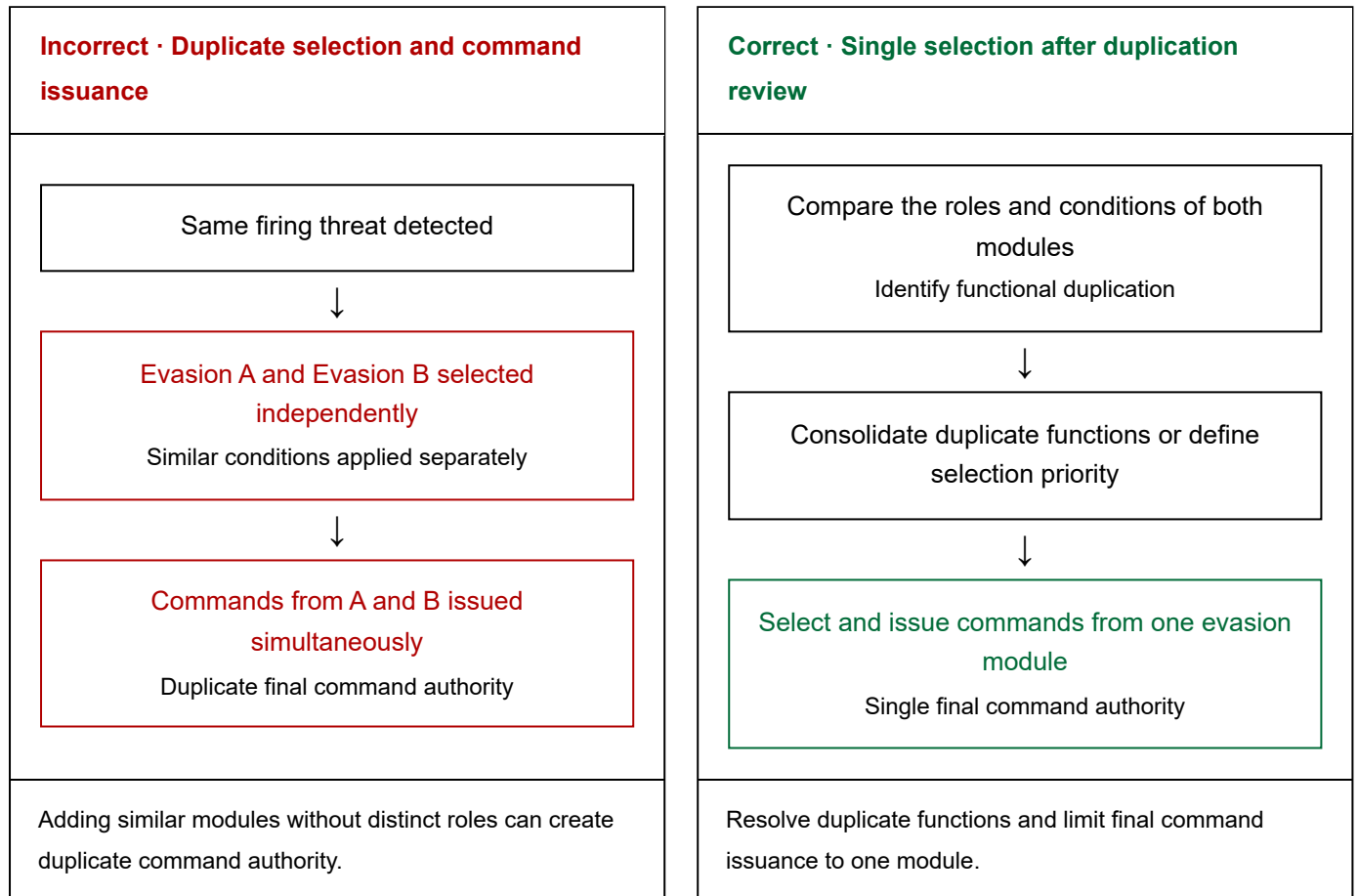


Figure 4-1. Single selection after duplication review

4-2 When the selected module changes, the previous module shall be verified to relinquish command authority.

Example scenario High-Yo-Yo ends and control transitions to precision tracking.

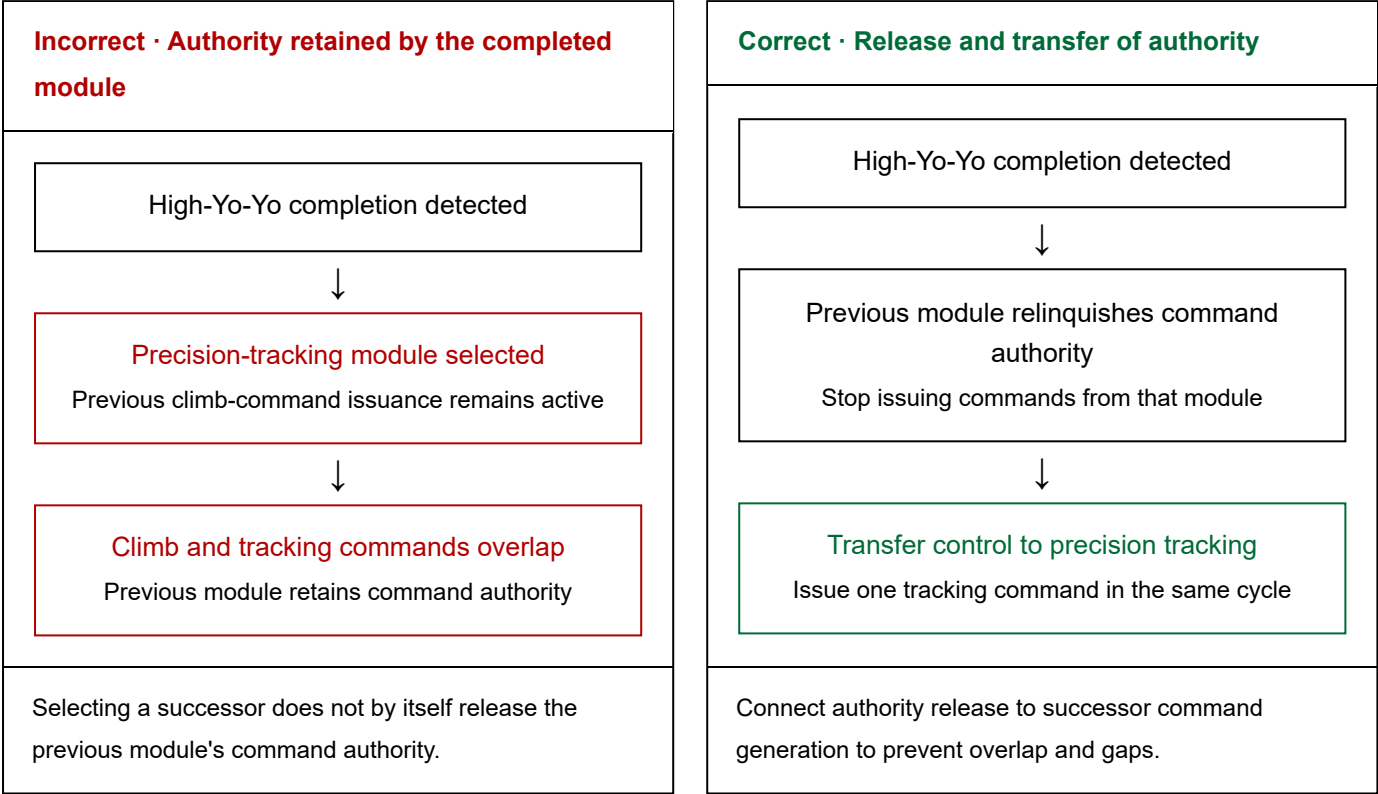


Figure 4-2. Release and transfer of authority

4-3 The safety module shall relinquish command authority when a safe state has been restored.

Example scenario The safety module's recovery-completion conditions are satisfied after terrain-collision avoidance.

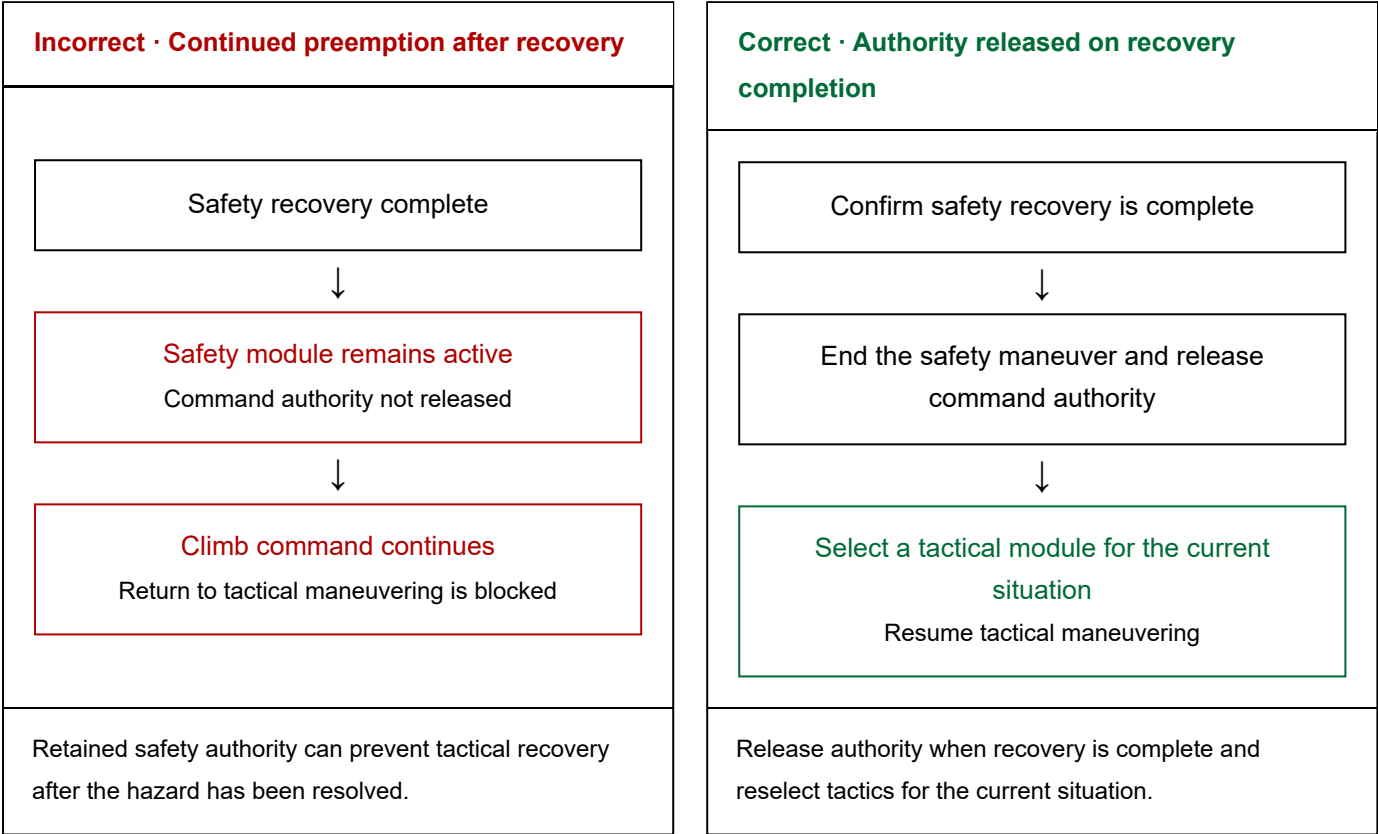


Figure 4-3. Authority released on recovery completion

05 Cyclic reevaluation

A Reactive BT shall reevaluate from the top of the tree in every control cycle.

- 5-1** **Guaranteed success through maneuver completion shall not be required as an entry prerequisite.**

Example scenario Evasion is currently required, but the opponent's future maneuvers cannot be determined.

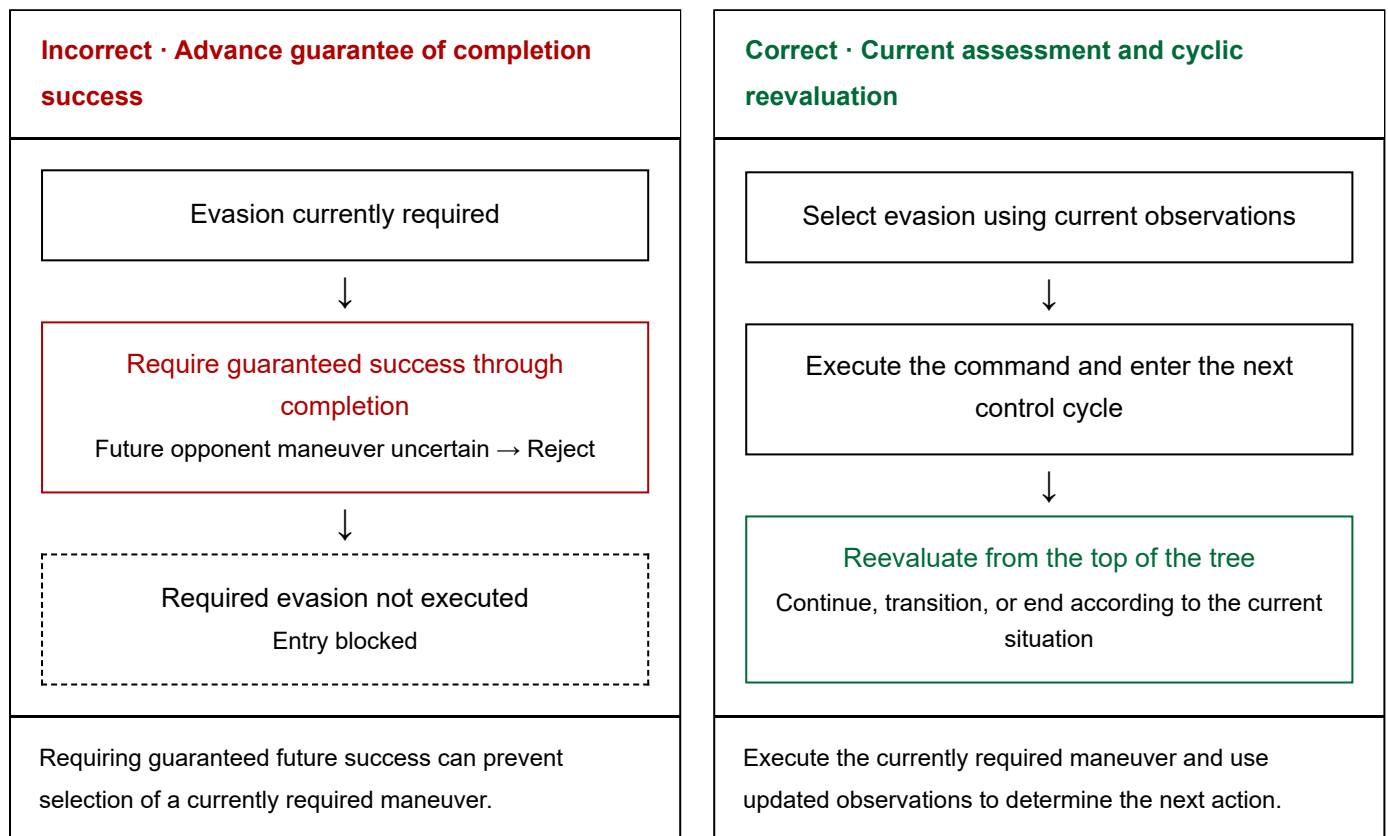


Figure 5-1. Current assessment and cyclic reevaluation

5-2 Excessively conservative entry conditions shall not prevent a required node from executing.

Example scenario A common threat assessment identifies a need for evasion, but additional observers are required to reconfirm the same threat.

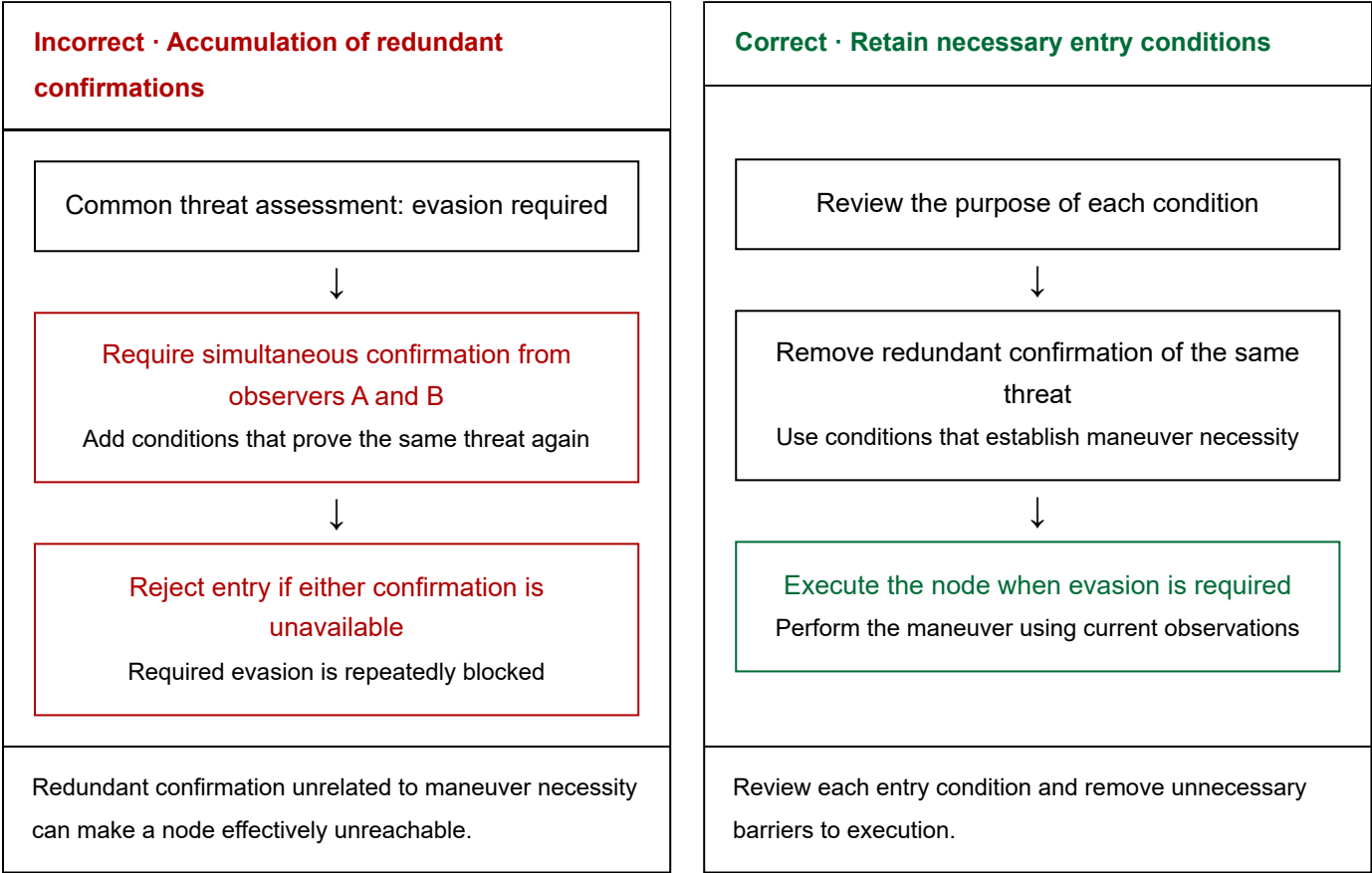


Figure 5-2. Retain necessary entry conditions